

SRINATH RAMALINGAM

📍 Boston, MA 📞 408-333-2604 ✉ htnaris@gmail.com  [LinkedIn Profile](#)  [Github Profile](#)

MACHINE LEARNING ENGINEER

Machine Learning Engineer with a proven track record in designing and deploying scalable AI models across NLP, computer vision, and recommender systems. Skilled in optimizing performance, building end-to-end pipelines, and leveraging big data to train neural networks. Excited to collaborate and drive solutions in highly performant teams.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Nearmap, San Francisco 05/2022-06/2024

- Reduced inference server pods by 50% and saved \$150,000 annually by optimizing model inference using Ziplt Model Merging, streamlining infrastructure without compromising performance or scalability.
- Designed and deployed a proactive hazard detection system using semantic segmentation models and geospatial analysis to predict risks like propane tank explosions, chimney obstructions, and roof ponding.
- Enhanced system reliability by refining model confidence prediction using temperature scaling and Bayesian optimization, reducing false positive rates by 20%, resulting in transparency in decision-making.
- Facilitated training and deployment of semantic segmentation models (U-Net, TransUNet) on remote sensing data, achieving sub-meter accuracy in property damage detection. Utilized distributed training on multi-GPUs.

Data Science Intern, App Orchid Inc., San Ramon 11/2021-05/2022

- Developed a sentiment analysis pipeline using OCR (Tesseract) and BERT to extract and classify text from old legal documents, achieving an 81% F1 score, a 5% improvement over baseline models.
- Automated sentiment analysis integration with existing legal review systems, reducing manual review time by 30% and enabling scalable handling of large document volumes.
- Enhanced the model's ability to interpret complex legal language by fine-tuning BERT on industry-specific datasets, improving sentiment analysis accuracy and relevance in legal document assessments.
- Established and deployed a robust data pipeline using Apache Airflow to automate quality checks, resolving over 1,000 labeling errors and significantly improving overall data quality and consistency.

PROJECTS

Netflix Insights Chatbot

- Developed a chatbot, leveraging RAG and Text-to-SQL techniques, to analyze Netflix Engagement Trends.
- Integrated vector Search, SQL agents and guardrails for intelligent querying, tool calling and augmentation to deliver reliable insights on viewing trends and title performance, enabling data-driven decision-making.

Large-Scale Training of Text-to-Speech Model

- Trained a GAN-based TTS model inspired by Microsoft's NaturalSpeech2, capable of generating audio in any speaker's voice from a prompt audio.
- Improved prosody with context-aware mechanisms using text vector embeddings.
- Trained an IPA-based phoneme recognition ASR model (WER: 4.2%, CER: 2.5%) and used its outputs to train an aligner network for precise pitch and duration predictions.

Fine-tuning LLaMA (LLM) with Parameter-Efficient Fine-Tuning (PEFT)

- Fine-tuned Llama 2 (7B) using LoRA and QLoRA for parameter-efficient adaptation, leveraging 4-bit quantization (NF4) and block-wise quantization to minimize memory usage without sacrificing performance.
- Utilized Hugging Face Transformers, PEFT libraries, and paged AdamW optimizers to streamline training, enabling scalable fine-tuning with mixed-precision and advanced scheduling techniques.

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OTHER IMPORTANT HIGHLIGHTS

ADDITIONAL EXPERIENCE

AutoML Machine Learning Engineer Intern, Telepath.io

Data Science Intern, Indian Institute of Technology

EDUCATION

Bachelor of Engineering, VIT University, Chennai, 07/2021

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI

AWS Certified Machine Learning - Specialty, Amazon

Machine Learning Engineer Professional Certificate: Google

TECHNICAL SKILLS

Languages	Python, Rust, Go, JavaScript, SQL, Bash
Frameworks	TensorFlow, PyTorch, CUDA, TensorRT, Flask, Django, LangChain, Llama Index
Distributed Systems	Hadoop, Hive, Spark, Kafka, Dask
MLOps	MLFlow, Kubeflow, Airflow, Docker, Kubernetes
Optimization	Quantization, Pruning, Model Distillation, Hyperparameter Tuning
Cloud Platforms	GCP, AWS, Azure, SageMaker, Vertex AI
CI/CD	Git, GitLab CI